Introduction to CG

- **Define Computer Graphics…**

  The technology associated with the use of computer technology to convert created or collected data into visual representations

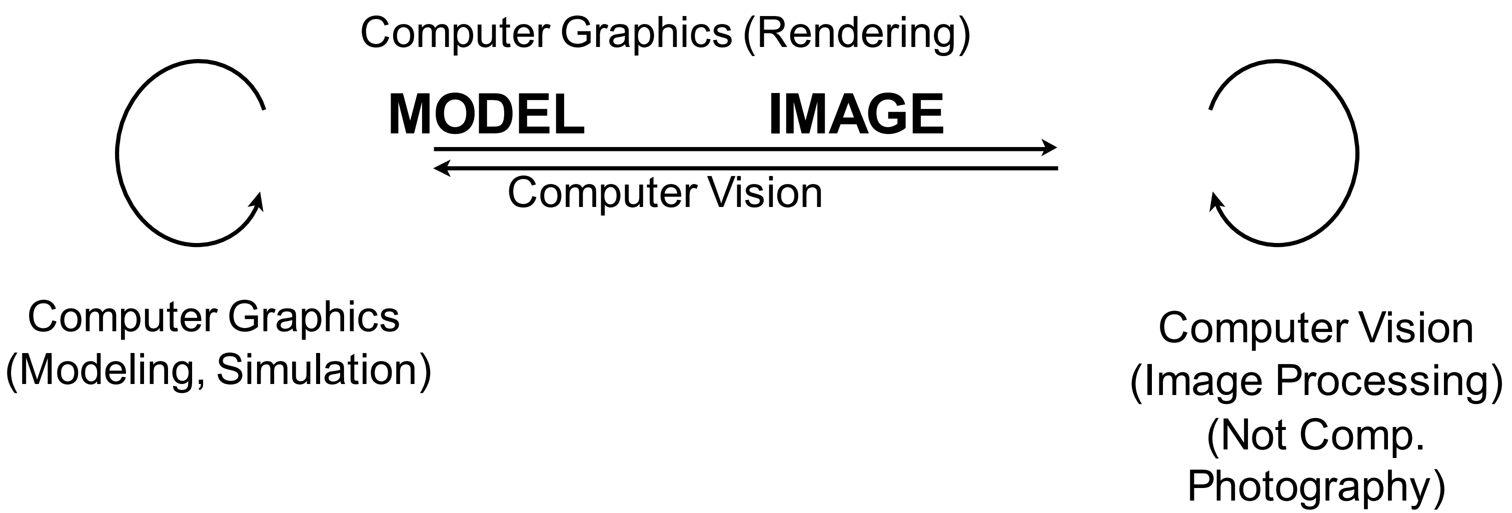
  Model

  Rendering → focus of course

  Display

Differences?

- Personal Understanding



- No clear boundaries

Graphics Process

Geometric Model

Scene Model

Surface Model

→

<u>Rendering</u>

transform, Light, Shade, & Rasterize

→

Image Display

Geometry Modeling

- There are many ways to describe geometry
  - Explicit geometry:

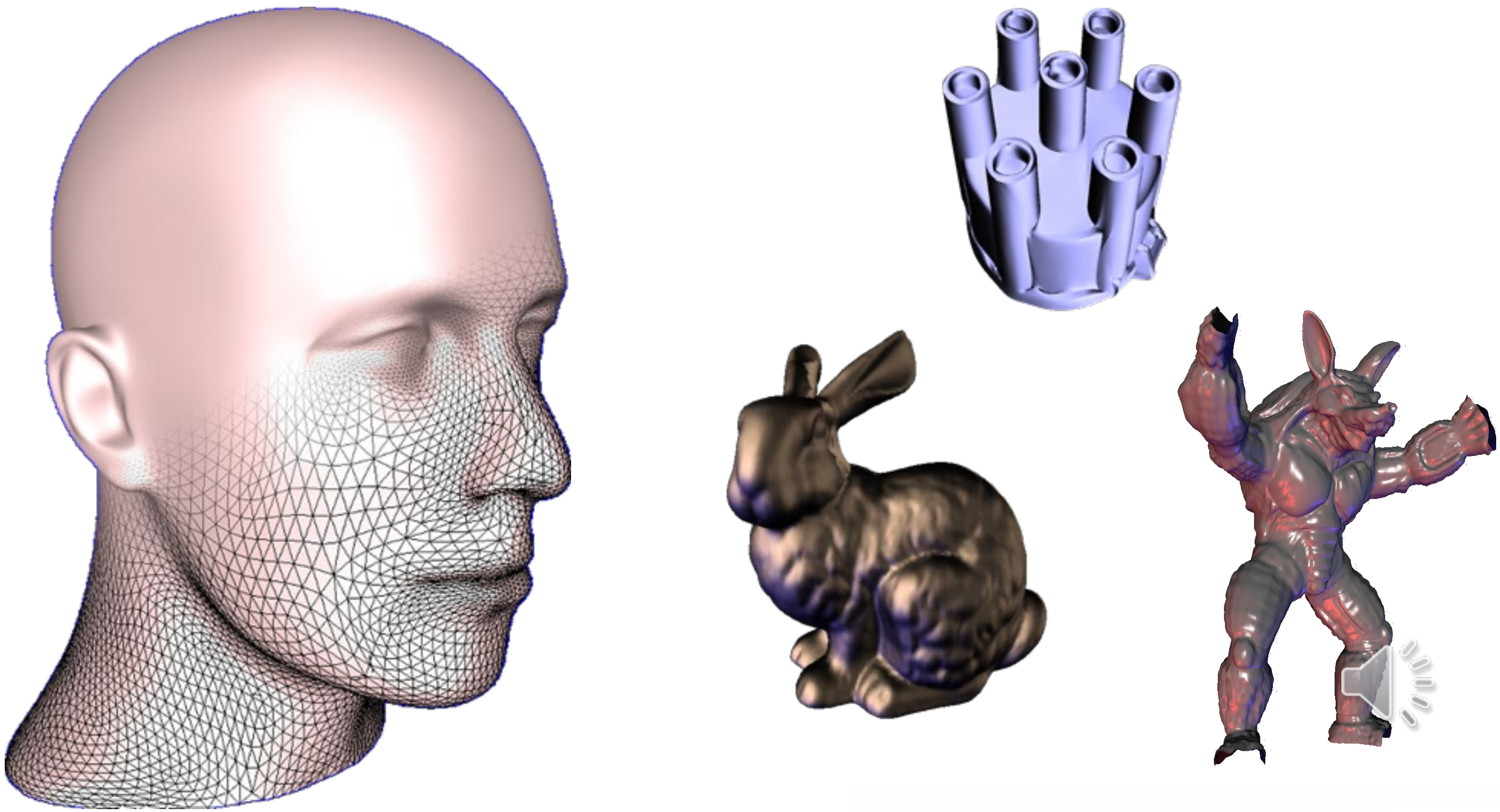
    Triangle meshes, Patches, Subdivision surfaces,…
  - Implicit geometry:

    Surface defined by $x^2 + y^2 + z^2 = 10$

    Fractal sets, procedural definition, …
  - Volume data

    Samples from MRI, ultra-sound, simulation…

Example of Triangle Meshes



Making Models

3D scanner

Model libraries

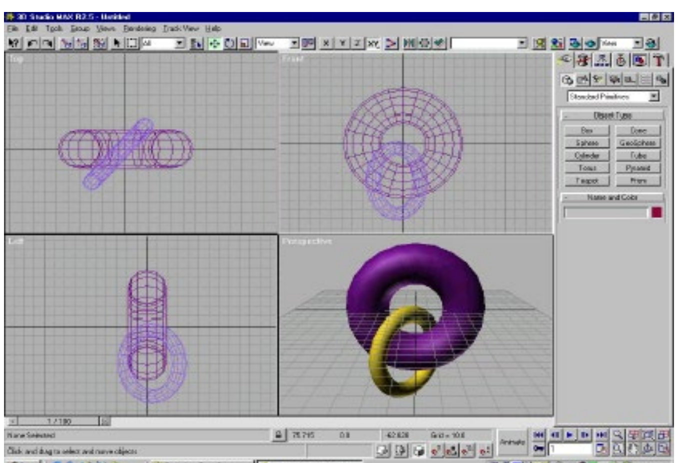
Interaction

Computer vision

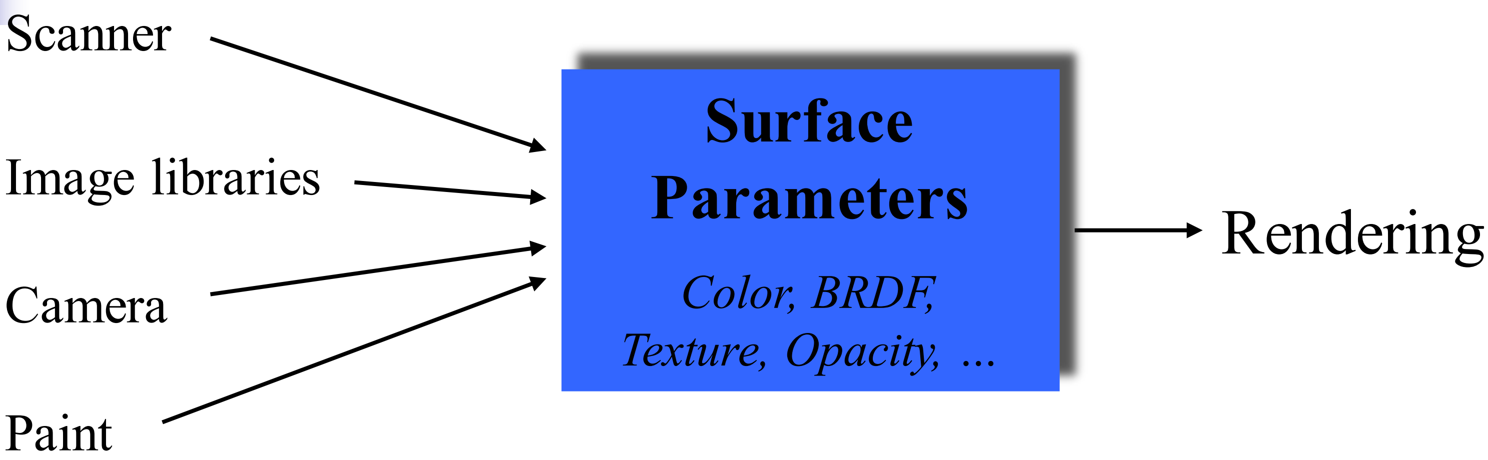
Geometric Modeling

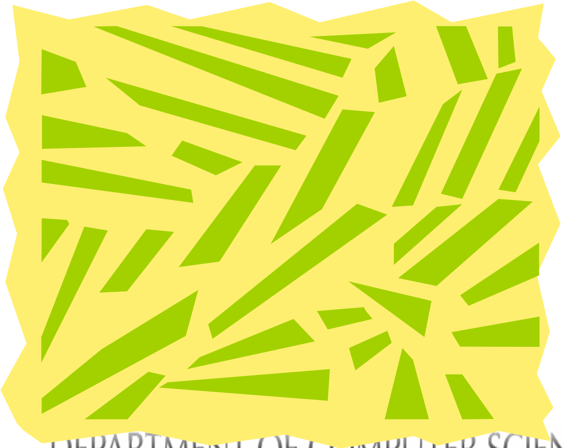
Points, Lines, Surfaces, …

Rendering



Making Surface Models





Rendering

Geometric Model

Surface Model

Rendering

Transformation

Image generation

IG = (lighting, shading, scan conversion)

Image Display

Image Display

Rendering → **Image Representation** → **Optical Modulation**

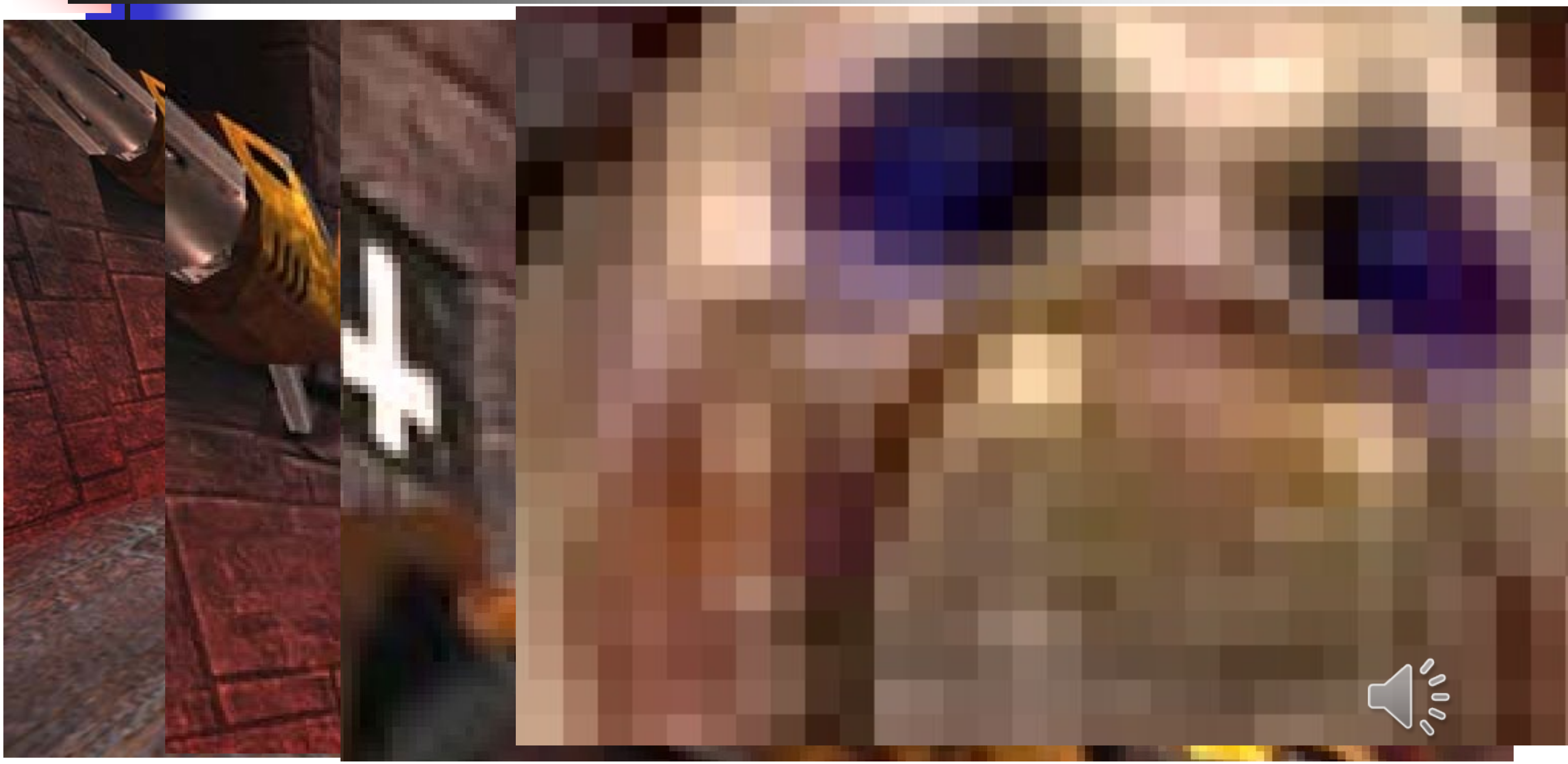
Image Representation

Pixel array,
Stroke list, NC cut list,
…

Optical Modulation

CRT, LED, LCD,
Plasma, Ink, Solid
material

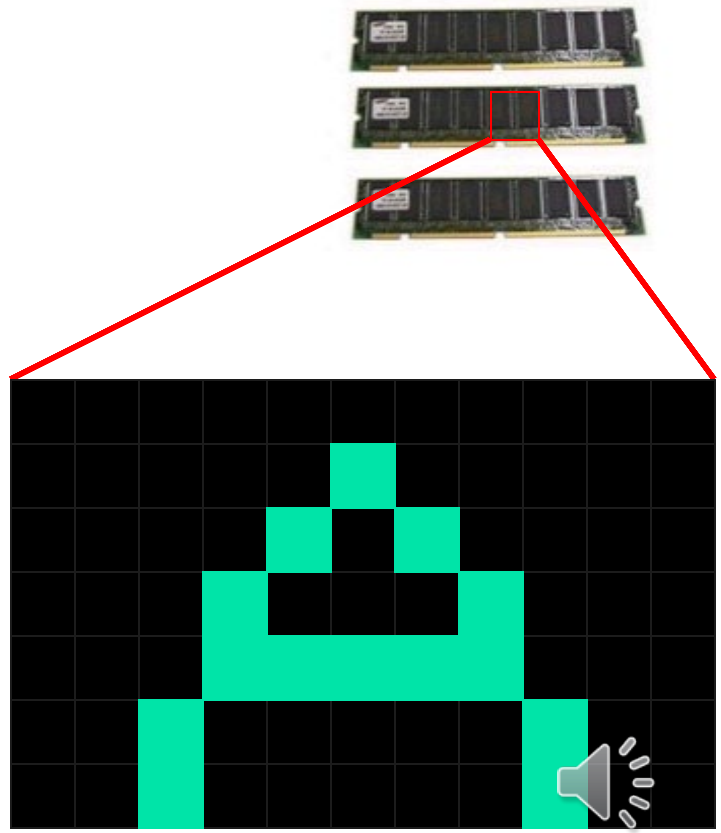
Digital Images: pixels



Frame Buffer

Frame Buffer

- A block of memory, dedicated that contains the pixel array to be passed to the optical display system
- Each pixel encodes color or other properties (e.g., opacity)



Frame Buffer Concepts

Pixel: One element of frame buffer
- uniquely accessible point in image

Resolution: Width x Height (in pixels)
- 640x480, 1280x1024, 1920x1080

Color depth: Number of bits per-pixel in the buffer
- 8, 16, 24, 32-bits for RGBA

Buffer size: Total memory allocated for buffer

Frame Buffer Opacity

Alpha

Used for compositing or merging images

Alpha channel – added to color

Holds the alpha value for every pixel

8 bit range: 0 (transparent) – 255 (opaque)

How Much Memory?

Buffer size = width * height *color depth

For example:

If width=640, height=480, color depth=24 bits

Buffer size = 640 * 480 * 3 = 921,600 bytes

If width=1920, height=1080, color depth=24 bits

Buffer size = 1920 * 1080 * 3 = 6,220,800 bytes

Display Device

- CRT (Cathode Ray Tube)
- LED (Light Emitting Diode)
- Plasma, Projection, HMD, Volumetric

Important Features:

size, resolution, field of view, pixel-pitch, color range, brightness, refresh-rate, black level, update mode (e.g., interlacing, …), distortion

Interaction

- Interaction is an important component of graphics applications
- Think about input devices in two ways:

  Physical device – that can be described by their real-world physical properties

  (mouse, keyboard, joystick…)

  Logical Device – application abstraction